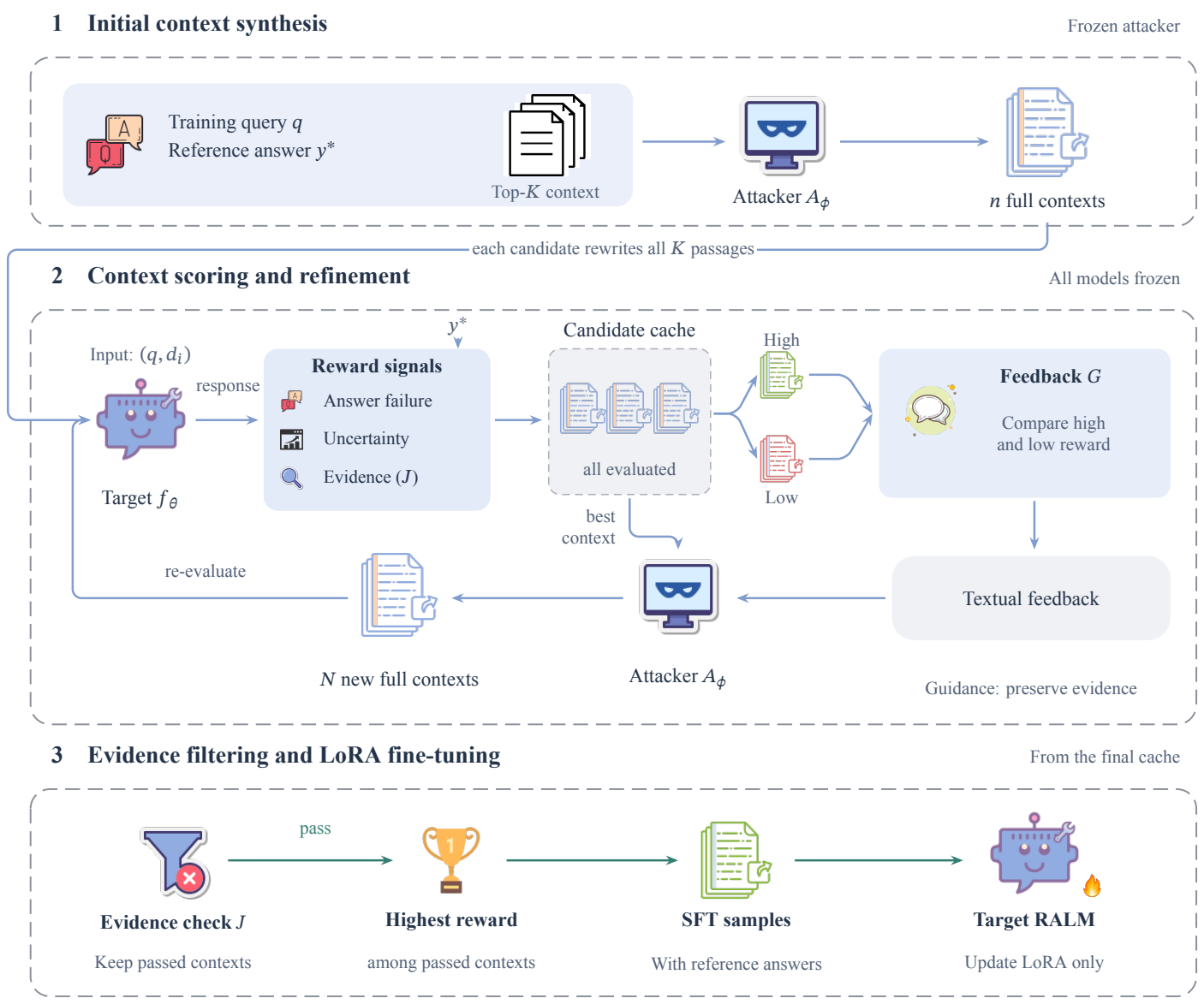

1 Initial context synthesis
Frozen attacker
Training query q
Reference answer y^*
Top-K context
Attacker A_ϕ
n full contexts
each candidate rewrites all K passages
2 Context scoring and refinement
All models frozen
Input: (q, d_i)
response
Target f_θ
y^*
Reward signals
Answer failure
Uncertainty
Evidence (J)
Candidate cache
all evaluated
High
Low
Feedback G
Compare high and low reward
best context
Textual feedback
Guidance: preserve evidence
Attacker A_ϕ
N new full contexts
re-evaluate
3 Evidence filtering and LoRA fine-tuning
From the final cache
Evidence check J
Keep passed contexts
pass
Highest reward
among passed contexts
SFT samples
With reference answers
Target RALM
Update LoRA only